SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION with Open CV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison.* (BTL 4)

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

Preprocessing and Feature Detection

Question:
A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.

Answer:
Different lighting changes pixel intensity and affects feature detection. Preprocessing techniques such as ***normalization, contrast enhancement, and noise removal*** *make images more consistent.*

Result: *Preprocessing improves the accuracy and reliability of feature detection.*

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2. Edge vs Corner Detection

Question:
In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching. Analyze why edge detection is insufficient and justify the use of corner detection in this case.

Answer:
Edge detection identifies boundaries but does not provide distinctive points for

matching. Corner detection identifies points where intensity changes in multiple directions.

Result: *Corner detection is more suitable for identifying keypoints used in image matching.*

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3. Hough Transform in Noisy Images

Question:
A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

Answer:
Noise creates false edges, which can produce incorrect lines in the Hough Transform. Applying **Gaussian or median filtering** *before detection reduces noise. Proper threshold and parameter selection also improves accuracy.*

Result: *Noise removal and parameter tuning improve line detection.*

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4. Feature Matching under Transformations

Question:
Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

Answer:
Traditional methods may depend on fixed feature sizes and orientations. Scale-invariant methods such as ***SIFT*** *detect features at different scales and orientations.*

Result: *Scale-invariant methods provide more reliable matching under size and rotation changes.*

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5. PCA for Feature Optimization

Question:
A system extracts a large number of features, leading to high computational cost and

redundancy. Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

Answer:
PCA transforms many correlated features into fewer principal components. It removes redundant information while retaining important variance.

Result: *PCA reduces computational cost, memory usage, and feature redundancy.*

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6. Preprocessing Impact on Edge Detection

Question:
A system performs edge detection on raw images containing noise and uneven illumination, resulting in fragmented edges. Analyze the problem and explain how preprocessing improves edge detection accuracy and continuity.

Answer:
Noise creates false edges and uneven illumination produces weak or inconsistent edges. **Noise filtering, normalization, and contrast enhancement** *improve the image before edge detection.*

Result: *Preprocessing produces cleaner and more continuous edges.*

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7. Corner Detection vs Edge Detection in Tracking

Question:
A tracking system uses edge detection but fails to track moving objects accurately across frames. Analyze why edges are insufficient and justify the use of corner detection for reliable tracking.

Answer:
Edges represent boundaries but may not provide unique points for tracking. Corners are distinctive points that can be identified across multiple frames.

Result: *Corner detection provides more reliable keypoints for object tracking.*

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8. Hough Transform for Shape Detection

Question:
A system uses Hough Transform to detect circular shapes but fails in complex backgrounds. Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

Answer:
Complex backgrounds create extra edges and false circles. ***Noise removal, edge detection, thresholding, and proper radius and accumulator parameters*** *can improve detection.*

Result: *Preprocessing and parameter tuning increase circle detection accuracy.*

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9. Feature Matching under Illumination Changes

Question:
Two images of the same scene are captured under different lighting conditions, causing feature matching errors. Analyze why matching fails and explain how preprocessing or robust descriptors improve performance.

Answer:
Lighting changes alter pixel intensities and feature descriptors. ***Histogram equalization, normalization, and robust descriptors such as SIFT*** *can reduce the effect of illumination changes.*

Result: *Preprocessing and robust descriptors improve matching reliability.*

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10. PCA in Face Recognition

Question:
A face recognition system processes high-dimensional image data, resulting in slow performance. Analyze how PCA helps in reducing dimensionality while preserving important features.

Answer:
Face images contain many pixels and therefore high-dimensional data. PCA converts them into fewer important components while retaining major facial information.

Result: *PCA reduces computation and storage while maintaining useful facial features.*

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11. Edge Detection in Low Contrast Images

Question:
An image has low contrast, causing edge detection algorithms to fail. Analyze the problem and explain how preprocessing enhances edge detection results.

Answer:
Low contrast produces small intensity differences between objects and backgrounds. ***Contrast enhancement and histogram equalization*** *increase intensity differences before edge detection.*

Result: *Preprocessing makes weak edges clearer and easier to detect.*

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12. Corner Detection in Textured Regions

Question:
A system applies corner detection in highly textured regions, resulting in excessive keypoints. Analyze the issue and suggest methods to improve feature selection.

Answer:
Highly textured regions contain many intensity changes, producing too many corners. Increasing the ***corner response threshold*** *and applying non-maximum suppression can remove weak points.*

Result: *Proper thresholding reduces unnecessary keypoints and improves efficiency.*

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13. Feature Matching with Noise

Question:
Feature matching between two images fails due to noise in one image. Analyze the impact of noise and explain how preprocessing improves matching accuracy.

Answer:
Noise changes pixel values and creates false or unstable features. Applying ***Gaussian or median filtering*** *before feature extraction reduces noise.*

Result: *Noise removal produces more stable features and improves matching accuracy.*

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14. PCA and Information Loss

Question:
A system applies PCA aggressively and loses important feature details. Analyze the trade-off between dimensionality reduction and information loss, and justify optimal component selection.

Answer:
Using too few principal components reduces computation but may remove important information. Using more components preserves information but increases computational cost.

Result: *Components should be selected based on the required* ***explained variance****, balancing efficiency and information preservation.*

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15. Combined Feature Detection and Matching

Question:
A system detects features correctly but fails during matching due to incorrect descriptor selection. Analyze the issue and explain how appropriate feature descriptors improve matching performance.

Answer:
Feature detection finds keypoints, while descriptors describe those keypoints for matching. An unsuitable descriptor may not represent features effectively. Using appropriate descriptors such as ***SIFT, ORB, or SURF*** *improves correspondence.*

Result: *Correct descriptor selection improves matching accuracy and reduces false matches.*

Code of Conduct:

I _____K.L.V.S KARTHIK(192424177)_______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.L.V.S KARTHIK(19242417)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand